

Infrastructure Case / HJB Numerical Methods

Session 27 • Track 1: Infrastructure asset case • Track 2: Finite-difference HJB solver

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

Graduate Finance Course • Spring 2027 • Session 27 of 32

Session 27: split-track agenda

TRACK 1 (Practitioner)

Infrastructure case workshop

- ▶ Case: Macquarie infrastructure fund — toll road project
- ▶ Long-horizon Jensen bias is largest here ($T=15-25$ yrs)
- ▶ Apply $\pi(L,T)$ term structure
- ▶ Compare DCF and GE-LAV NPVs across regimes
- ▶ Deliverable: 2-page case memo

TRACK 2 (Researcher)

HJB numerical methods

- ▶ Finite-difference discretization of HJB equation
- ▶ Implicit vs explicit schemes; stability analysis
- ▶ Free-boundary detection algorithms
- ▶ Convergence rates
- ▶ Problem: code a simple HJB solver in Python

Session 27 summary

What we accomplished today

- 1 Session 27 runs split-track: practitioners and researchers work on different deliverables
- 2 Track 1: case-based, applied analysis with deliverable IC memo
- 3 Track 2: theoretical, with deliverable problem set or proof
- 4 Both tracks reconvene in Session 32 for final project presentations

Next session

Session 28: continued split-track or capstone